

Functions: Modulus & Inverse

ANSWER KEY



Evaluating and graphing modulus functions

- Given $f(x) = |2x - 3|$, find $f(1)$ and $f(-2)$.
 $f(1) = |2(1) - 3| = |-1| = 1$. $f(-2) = |2(-2) - 3| = |-4 - 3| = |-7| = 7$.
- The graph shows $g(x) = |x + 4| - 2$. State the coordinates of its vertex and the range of g .
The vertex occurs where $x + 4 = 0$, i.e. $x = -4$, giving $g(-4) = 0 - 2 = -2$. Vertex: $(-4, -2)$. Since $|x + 4| \geq 0$ for all x , the range is $g(x) \geq -2$.

Solving modulus equations

- Solve $|3x - 2| = 7$.
Case 1: $3x - 2 = 7 \Rightarrow x = 3$. Case 2: $3x - 2 = -7 \Rightarrow x = -\frac{5}{3}$. Solutions: $x = 3$ or $x = -\frac{5}{3}$.
- Solve $|x + 5| = 2x - 1$, rejecting any extraneous solution.
Case 1: $x + 5 = 2x - 1 \Rightarrow x = 6$. Check: $2(6) - 1 = 11 \geq 0$ and $|11| = 11$ ✓. Case 2:
 $-(x + 5) = 2x - 1 \Rightarrow -3x = 4 \Rightarrow x = -\frac{4}{3}$. Check: $2\left(-\frac{4}{3}\right) - 1 = -\frac{11}{3} < 0$, which cannot equal a modulus (always ≥ 0), so this is extraneous. Only solution: $x = 6$.
- Solve $|4 - x| = 3x$, rejecting any extraneous solution.
Case 1: $4 - x = 3x \Rightarrow 4 = 4x \Rightarrow x = 1$. Check: $3(1) = 3 \geq 0$ ✓. Case 2:
 $-(4 - x) = 3x \Rightarrow x - 4 = 3x \Rightarrow -4 = 2x \Rightarrow x = -2$. Check: $3(-2) = -6 < 0$, invalid since a modulus cannot equal a negative number. Only solution: $x = 1$.

Solving modulus inequalities

- Solve $|2x - 1| < 5$.
 $-5 < 2x - 1 < 5 \Rightarrow -4 < 2x < 6 \Rightarrow -2 < x < 3$.
- Solve $|x - 3| \geq 4$.
 $x - 3 \geq 4$ or $x - 3 \leq -4$, giving $x \geq 7$ or $x \leq -1$.
- Solve $3 - |x + 2| > -1$.
Subtract 3: $-|x + 2| > -4$. Multiply by -1 and reverse the inequality: $|x + 2| < 4$. So $-4 < x + 2 < 4$, giving $-6 < x < 2$.

Modulus and graph transformations

- 9 The graph shows $y = |x^2 - 4x|$. Explain why the parts of the original parabola $y = x^2 - 4x$ between $x = 0$ and $x = 4$ appear reflected upward.
- $y = x^2 - 4x = x(x - 4)$ is negative for $0 < x < 4$ (the parabola dips below the x -axis between its two roots). Taking the modulus reflects any portion of the graph where the original function is negative upward across the x -axis, so that dipped section is flipped to lie above the axis, while the parts where $x^2 - 4x \geq 0$ are unchanged.
- 10 Explain the difference between the graphs of $y = |f(x)|$ and $y = f(|x|)$, using $f(x) = x - 2$ as an example.
- $y = |f(x)| = |x - 2|$ takes the graph of $y = x - 2$ and reflects only the part that lies below the x -axis (where $x < 2$) upward, leaving the part where $x \geq 2$ unchanged. $y = f(|x|) = |x| - 2$ instead replaces the graph for $x < 0$ with a mirror image of the $x > 0$ portion reflected across the y -axis, discarding the original left-hand definition of f entirely and producing a graph symmetric about the y -axis.

Finding inverse functions

- 11 Find $f^{-1}(x)$ for $f(x) = 3x + 2$.
- Let $y = 3x + 2$. Solve for x : $x = \frac{y - 2}{3}$. So $f^{-1}(x) = \frac{x - 2}{3}$.
- 12 Find $f^{-1}(x)$ for $f(x) = \frac{2x + 1}{x - 3}$, $x \neq 3$.
- Let $v = \frac{2x + 1}{x - 3}$. Then $y(x - 3) = 2x + 1 \Rightarrow yx - 3y = 2x + 1 \Rightarrow x(y - 2) = 3y + 1 \Rightarrow x = \frac{3y + 1}{y - 2}$. So $f^{-1}(x) = \frac{3x + 1}{x - 2}$.

Restricting the domain for invertibility

- 13 Explain why $g(x) = x^2 - 6x + 5$ (domain all reals) is not one-to-one, and state a suitable restricted domain that makes g invertible.
- $g(x) = (x - 3)^2 - 4$ is a parabola symmetric about $x = 3$, so two different x -values equidistant from 3 give the same output (e.g. $g(2) = g(4) = -3$), meaning g is not one-to-one over all reals. Restricting the domain to $x \geq 3$ (where g is strictly increasing) makes g one-to-one and hence invertible.
- 14 Using the restricted domain $x \geq 3$ from the previous problem, find $g^{-1}(x)$ for $g(x) = (x - 3)^2 - 4$, and state its domain.
- Let $y = (x - 3)^2 - 4$, so $y + 4 = (x - 3)^2$. Since $x \geq 3$, take the positive root: $x - 3 = \sqrt{y + 4}$, so $x = 3 + \sqrt{y + 4}$. Thus $g^{-1}(x) = 3 + \sqrt{x + 4}$, with domain $x \geq -4$ (matching the range of g).
- 15 For $h(x) = (x + 2)^2$ restricted to $x \leq -2$, find $h^{-1}(x)$ and state its domain.
- Let $y = (x + 2)^2$. Since $x \leq -2$, $x + 2 \leq 0$, so take the negative root: $x + 2 = -\sqrt{y}$, giving $x = -2 - \sqrt{y}$. Thus $h^{-1}(x) = -2 - \sqrt{x}$, with domain $x \geq 0$ (matching the range of h).

Composite functions with modulus

- 16 Given $f(x) = |x - 1|$ and $g(x) = 2x + 3$, find $fg(x)$ in simplified form, and evaluate $fg(0)$.
 $fg(x) = f(g(x)) = |(2x + 3) - 1| = |2x + 2| = 2|x + 1|$. $fg(0) = 2|1| = 2$.
- 17 Using the same $f(x) = |x - 1|$ and $g(x) = 2x + 3$, find $gf(x)$, and evaluate $gf(-4)$.
 $gf(x) = g(f(x)) = 2|x - 1| + 3$. $gf(-4) = 2|-5| + 3 = 2(5) + 3 = 13$.
- 18 Using $fg(x) = 2|x + 1|$ found earlier, solve $fg(x) = 5$.
 $2|x + 1| = 5 \Rightarrow |x + 1| = 2.5$. So $x + 1 = 2.5 \Rightarrow x = 1.5$, or $x + 1 = -2.5 \Rightarrow x = -3.5$.

Combining modulus with quadratics and other functions

- 19 Solve $|x^2 - 9| = 7$.
 Case 1: $x^2 - 9 = 7 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$. Case 2: $x^2 - 9 = -7 \Rightarrow x^2 = 2 \Rightarrow x = \pm\sqrt{2}$. All four solutions:
 $x = 4, -4, \sqrt{2}, -\sqrt{2}$.
- 20 Solve $|2x - 5| = |x + 1|$ by squaring both sides.
 $(2x - 5)^2 = (x + 1)^2 \Rightarrow 4x^2 - 20x + 25 = x^2 + 2x + 1 \Rightarrow 3x^2 - 22x + 24 = 0$. By the quadratic formula:
 $x = \frac{22 \pm \sqrt{484 - 288}}{6} = \frac{22 \pm \sqrt{196}}{6} = \frac{22 \pm 14}{6}$. So $x = 6$ or $x = \frac{4}{3}$. Both check: at $x = 6$, $|7| = |7|$; at $x = \frac{4}{3}$, $|\frac{2}{3}| = |\frac{7}{3}|$.
- 21 Solve $|2x + 1| = x + 4$, checking both solutions are valid.
 Case 1: $2x + 1 = x + 4 \Rightarrow x = 3$. Check: $x + 4 = 7 \geq 0$ ✓, and $|7| = 7$ ✓. Case 2:
 $-(2x + 1) = x + 4 \Rightarrow -3x = 5 \Rightarrow x = -\frac{5}{3}$. Check: $x + 4 = \frac{7}{3} \geq 0$ ✓, and $|\frac{10}{3} + 1| = |-\frac{7}{3}| = \frac{7}{3}$ ✓.
 Both solutions are valid: $x = 3$ or $x = -\frac{5}{3}$.

A well-designed word problem: calibrating a sensor

- 22 This question has three parts. An engineer models the deviation of a sensor reading from its calibrated target as $d(x) = |3x - 12|$, where x is the applied voltage (volts) and $d(x)$ is the deviation (millivolts). (a) Find the voltage at which the deviation is zero. (b) The sensor is "within tolerance" when $d(x) \leq 6$. Solve this inequality to find the acceptable range of voltages. (c) For $x \geq 4$, define the calibration function $c(x) = d(x) + 2x - 5$. Explain why c is invertible on this restricted domain, and find $c^{-1}(x)$.
- (a) $|3x - 12| = 0 \Rightarrow 3x - 12 = 0 \Rightarrow x = 4$ volts. (b) $|3x - 12| \leq 6 \Rightarrow -6 \leq 3x - 12 \leq 6 \Rightarrow 6 \leq 3x \leq 18 \Rightarrow 2 \leq x \leq 6$. (c) For $x \geq 4$, $3x - 12 \geq 0$, so $d(x) = 3x - 12$ without needing the modulus sign flip. Then $c(x) = (3x - 12) + 2x - 5 = 5x - 17$, which is linear with positive slope 5, hence strictly increasing and one-to-one, so c is invertible on $x \geq 4$. Let $y = 5x - 17$: $x = \frac{y + 17}{5}$. So $c^{-1}(x) = \frac{x + 17}{5}$, with domain $x \geq 3$ (since $c(4) = 20 - 17 = 3$).